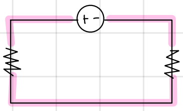
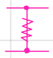
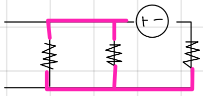
NODAL ANALYSIS



- identify nodes
- use node attatched to 3/more branches / lines

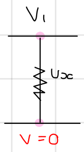
NB





- usable nodes : attatched 3

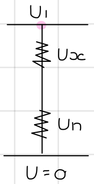
* reason: place where current splits



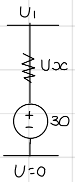
V1 is sum of all voltages from point where current split to where it splits again.

$U_1 = U_x$

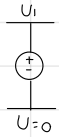
choose bottom node as ground



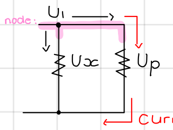
$U_1 = U_x + U_n$



$U_1 = U_x + 30$



$U_1 = 30$

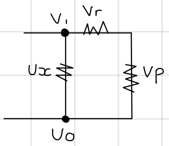


$U_1 = U_x$

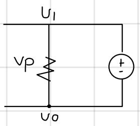
$U_1 = U_p$ $\Rightarrow U_x = U_p$

current does not split

if 2/more branches share 2 nodes, they are //
↳ // same voltage



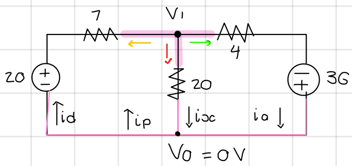
$V_1 = V_R + V_p = U_x$



$U_1 = 30 = U_p$

nodal analysis steps

1. Identify nodes : connected 3/more lines / branches
2. Assign variables & 1 node as ground (U=0)
3. check for supernodes and if → take it out by using supernode equation, then combine the remaining nodes
4. Assume current is leaving the node, unless otherwise stated (original current source opposing assumed direction
5. Apply nodal analysis



$V_0 = 0\,V$

note: assume current leaving even if they have arrows opposite direction (not from current source)

$I = \frac{V_{diff}}{R_{total}}$

voltage source / enter - polarity

$$\boxed{\frac{V_1-(+20)-0}{7}} + \boxed{\frac{V_1-(0)-0}{20}} + \boxed{\frac{V_1-(-36)-0}{4}} = 0$$

current from V1 split again at U0

$V_1 = -13{,}87\ V$

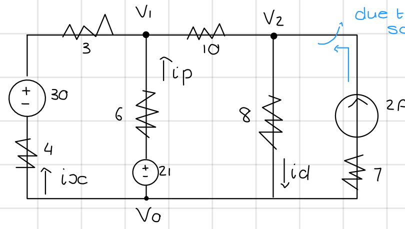
$i_x : \boxed{\frac{V_1}{20}} = -0{,}69$

opposing assume direction

$i_p : -\boxed{\left(\frac{V_1}{20}\right)} = 0{,}69$

$i_d : -\boxed{\left(\frac{V_1-20}{7}\right)} = 4{,}84$

$i_a : \boxed{\frac{V_1+36}{4}} = 5{,}53$



due to current source

$$V_1 : \frac{V_1-(30)-0}{3+4} + \frac{V_1-(21)-0}{6} + \frac{V_1-(0)-V_2}{10} = 0$$

$$\frac{V_1}{7} - \frac{30}{7} + \frac{V_1}{6} - \frac{21}{6} + \frac{V_1}{10} - \frac{V_2}{10} = 0$$

$$V_1\left(\frac{1}{7}+\frac{1}{6}+\frac{1}{10}\right) + V_2\left(-\frac{1}{10}\right) = \frac{30}{7} + \frac{21}{6}$$

$$V_1\left(\frac{43}{105}\right) + V_2\left(-\frac{1}{10}\right) = \frac{109}{14}$$

opposing chosen direction

$$V_2 : \frac{V_2-(0)-V_1}{10} + \frac{V_2-(0)-0}{8} - 2 = 0$$

current source

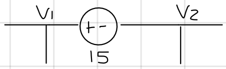
$$-\frac{1}{10}V_1 + \frac{9}{40}V_2 = 2$$

$V_1 = 23.76\ V$ $V_2 = 19.45\ V$

$i_x : -\left(\frac{V_1-30}{7}\right) = 0{,}89\ A$ $i_p : -\left(\frac{V_1-21}{6}\right) = 0{,}39\ A$

$i_d : \frac{V_2}{8} = 2{,}43\ A$

SUPERNODE: voltage source between 2 node voltages without a resistor being involved



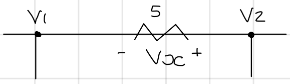
$V_1 - V_2 = 15$

when you see a supernode you take it out by using an equation

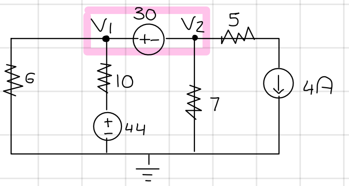
equation

node voltage close to + polarity of middle voltage
− node voltage close to − polarity of middle voltage
=
middle voltage

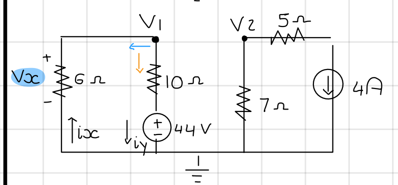
imaginary supernode:



$V_2 - V_1 = 5$



$V_1 - V_2 = 30$



$$\boxed{\frac{V_1}{6}} + \boxed{\frac{V_1-44}{10}} + \frac{V_2}{7} + 4 = 0$$

$$\frac{4}{15}V_1 + \frac{1}{7}V_2 = \frac{2}{5}$$

node 1 + node 2 = 0

$V_1 = 11.44$ $V_2 = -18.56$

Current cases

- $i_x = -\left(\frac{V_1}{6}\right) = -1{,}907$
- $i_y = \frac{V_1-44}{10} = -3{,}256$